United States International University-Africa

Education to take you places

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TASK: PROGRESS REPORT 2

1. Introduction

1.1 Project Overview

The **Radar! Underground Activism Hub** is a secure, peer-to-peer (P2P) communication platform designed to facilitate private, censorship-resistant, and decentralized communication for activists and community organizers. By leveraging advanced technologies such as WebRTC, IPFS, and community-led governance models, Radar aims to provide a robust alternative to traditional messaging platforms that often rely on centralized servers.

1.2 Motivation & Problem Statement

Traditional communication platforms, including popular applications like WhatsApp, Telegram, and Signal, depend on centralized infrastructure. This centralization poses significant risks, such as government surveillance, corporate influence, and data breaches. Even with encryption measures in place, these platforms often store metadata and user data, increasing the risk of compromised security and unauthorized access. Radar seeks to mitigate these vulnerabilities by employing a decentralized, ephemeral messaging approach, ensuring that no third party can access or retain sensitive user data.

1.3 Project Goals

The primary objectives of the Radar! Underground Activism Hub are:

- **Implementing a Fully Decentralized P2P Communication System:** Eliminating reliance on central servers to enhance privacy and security.
- **Developing a Secure Environment for Activist Communication:** Ensuring anonymity and protection against surveillance.
- **Enabling Real-Time Multimedia Communication:** Supporting messaging, voice, video, and file sharing with robust encryption.
- **Supporting Public and Private Rooms:** Facilitating different levels of community engagement.
- **Introducing Community-Based Moderation and Fact-Checking:** Preventing misinformation through decentralized control mechanisms.
- **Ensuring Metadata Protection and Privacy-Preserving Design:** Implementing ephemeral messaging and automatic peer verification via client-side cryptography.

2. System Design & Architecture

2.1 Justification for Web Mesh Architecture

Adopting a web mesh architecture offers several advantages over traditional centralized systems:

- **Enhanced Privacy:** By enabling direct browser-to-browser connections, web meshes ensure that data transmission occurs without intermediaries, reducing the risk of data interception or misuse.
- **Resilience Against Censorship:** Decentralized networks are inherently more resistant to censorship attempts, as there is no central point of control that can be targeted or shut down.
- **User Empowerment:** Web meshes distribute control among users, aligning with the principles of digital autonomy and reducing dependence on large tech corporations.

These benefits align with the goals of Radar, providing a secure and autonomous communication platform for activists.

2.2 Core Features

Radar integrates a range of privacy-enhancing features, including:

- **Multiple Peers Per Room:** Limited only by browser WebRTC connection limits.
- **Public & Private Rooms:** Public rooms allow open discussions, while private rooms require invitations.
- **Video & Audio Chat:** Secure, real-time multimedia communication.
- **Screen Sharing:** Allows users to share their screens securely.
- **Direct Messaging (DMs):** One-on-one encrypted conversations.
- **File Sharing:** Encrypted transfers with no size limits, using the room name as the encryption key.
- **Markdown & Syntax Highlighting:** Supports formatted messages, including code snippets.
- **Conversation Backfilling:** New participants receive past messages from active peers.
- **Multiline Messaging:** Supports multiline messages using Shift + Enter.
- **Dark & Light Themes:** Customizable UI for accessibility.
- **Automatic Peer Verification:** Utilizes client-side public-key cryptography for identity verification.

2.3 Anti-Features (Deliberate Privacy-Enhancing Exclusions)

To prioritize user privacy, Radar intentionally excludes:

- **Message Storage:** Messages are never persisted to disk; they disappear when users leave.
- **Central API Server:** The platform operates without a central server, ensuring decentralization.
- **Tracking, Analytics, or Telemetry:** No user data is collected, preserving anonymity.
- **Monetization or Corporate Influence:** The project remains community-driven and non-commercial.

3. Moderation & Fact-Checking

3.1 Community-Based Moderation

Radar plans to employ a decentralized moderation system, allowing communities to self-govern without reliance on a central authority. This approach aligns with initiatives like Reddit's content moderation tools, which aim to enhance user engagement through community-driven efforts.

The moderation framework consists of:

- **Flagging System:** Users can report inappropriate content.
- **Voting-Based Content Removal:** Community members collectively decide on flagged content.
- **Moderator Roles:** Trusted members can be assigned as moderators within rooms.
- **Decentralized Decision-Making:** No single entity has the power to control or censor content.

3.2 Community-Based Fact-Checking

Misinformation is managed through collective verification by community members, similar to platforms like X's Community Notes and initiatives like Verificado 2018. This approach includes:

- **Fact-Checking Reputation System:** Users who consistently verify accurate information gain credibility.
- **Flagging Misinformation:** Community members can highlight false claims.
- **Peer Consensus Model:** Content is reviewed by multiple users before being marked as misleading.

3.3 Case Studies Supporting Community-Based Fact-Checking and Moderation

Several real-world initiatives demonstrate the effectiveness of community-based moderation and fact-checking:

- **Africa Check:** A leading independent fact-checking organization that works with journalists and researchers to debunk misinformation across the African continent.
- **ProPublica's Collaborative Journalism Model:** Engages readers and contributors in identifying and verifying information to hold institutions accountable.
- **Reddit's Community-Driven Moderation:** Uses a decentralized system where subreddit moderators enforce community rules, demonstrating scalable self-governance.

These models provide a framework for Radar's approach to ensuring content integrity while preserving user autonomy.

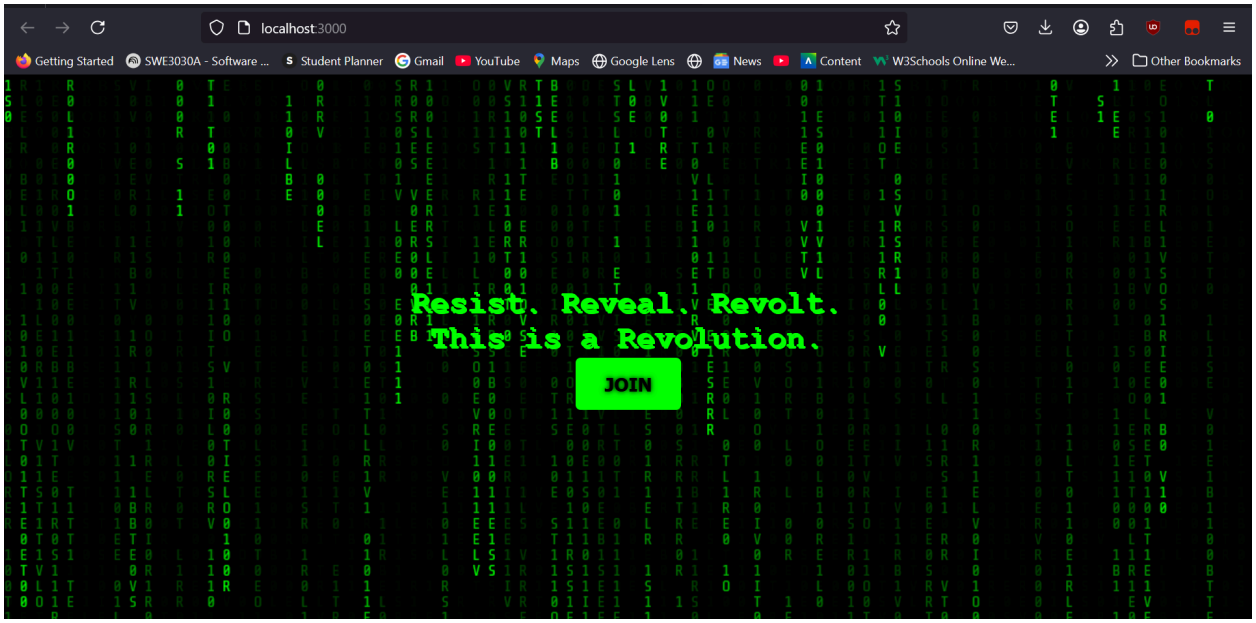
4. Implementation Progress & Challenges

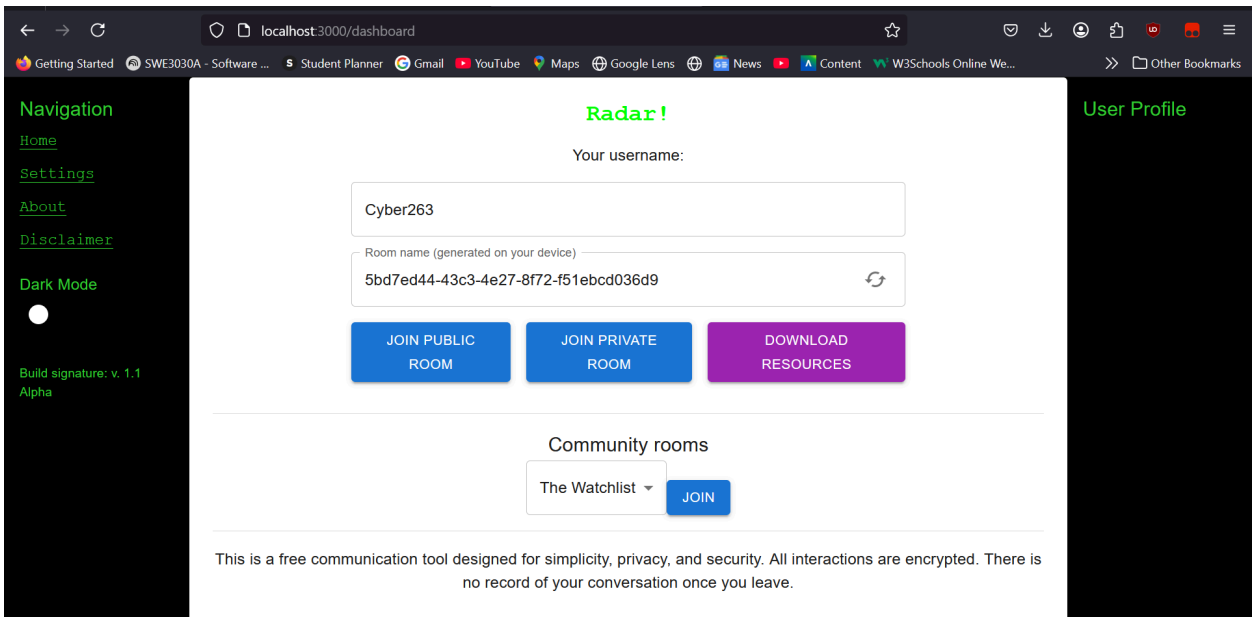
4.1 Current Progress (with Screenshots)

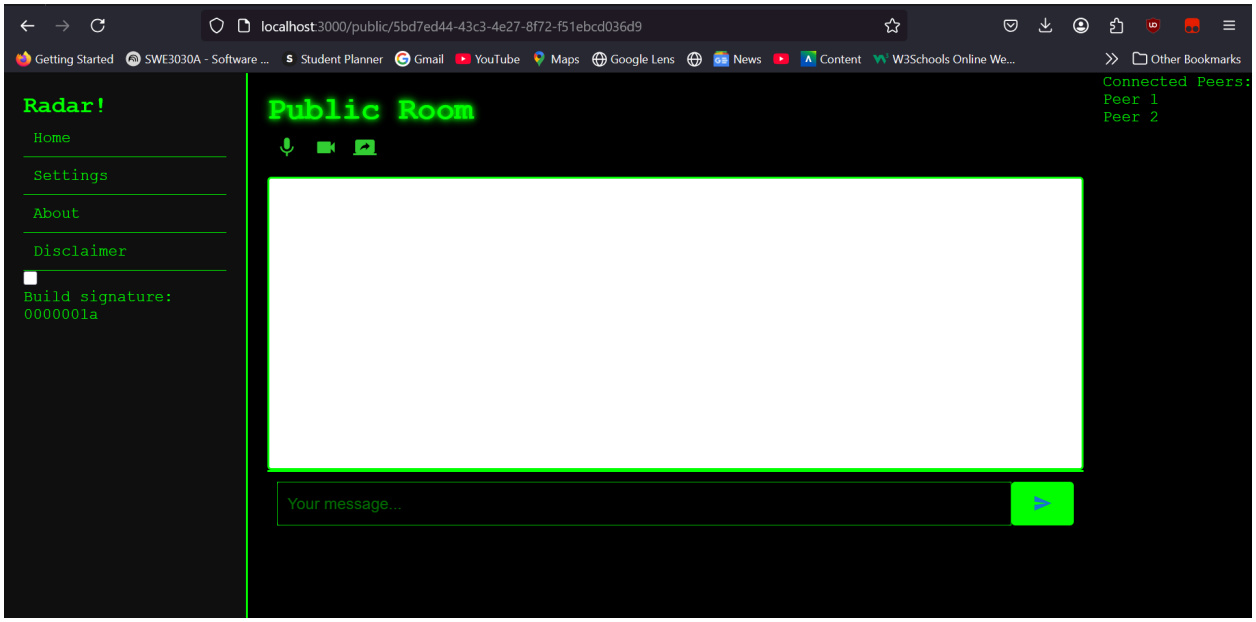
As of the latest assessment, the Radar! Underground Activism Hub has achieved approximately 60-65% completion. Below are the milestones achieved, accompanied by screenshots showcasing the platform's development:

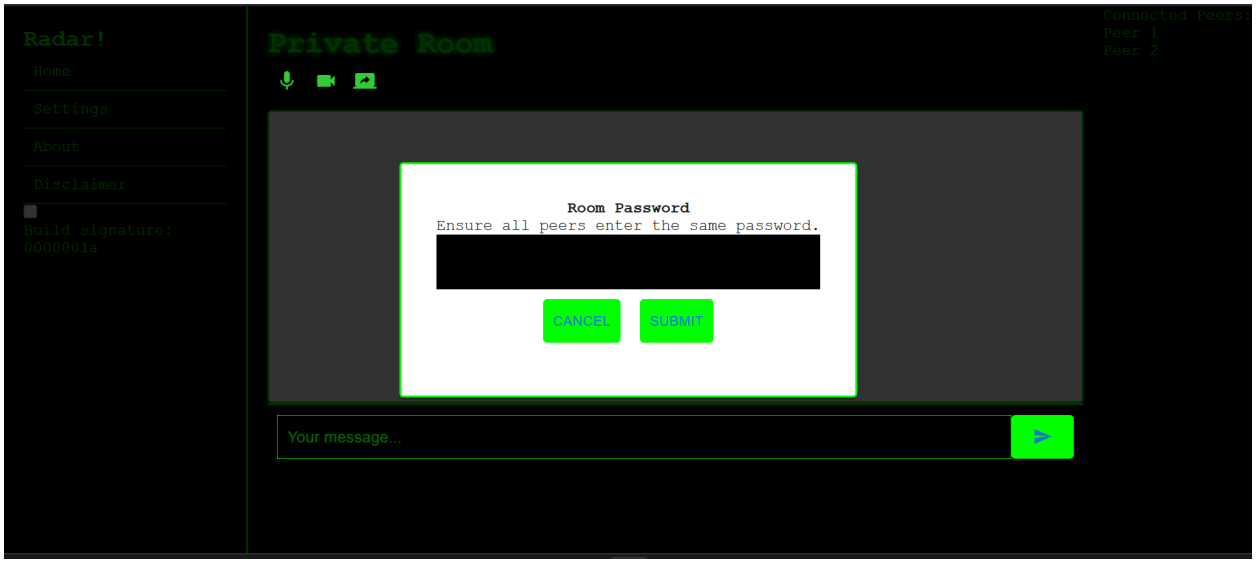
- **Core Communication Features:** Secure messaging, audio/video calls, and file sharing.
- **Public and Private Rooms:** Functional room structures for open and restricted discussions.
- **Basic Community Moderation Tools:** Initial implementation of flagging and reporting mechanisms.
- **Web Mesh Architecture:** Development of decentralized network infrastructure.

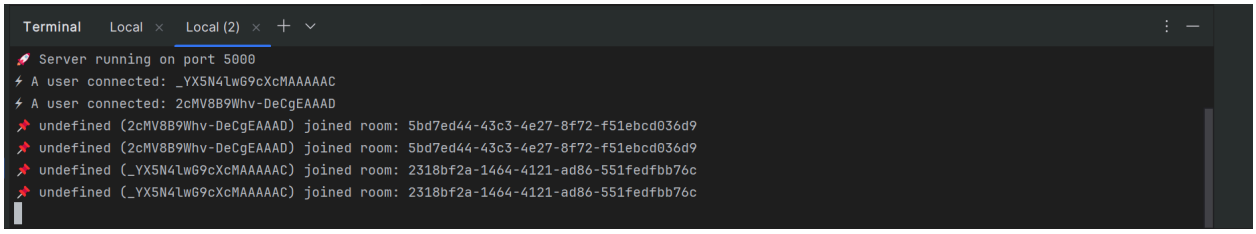
4.1.0. Screenshots






localhost:3000/public/5bd7ed44-43c3-4e27-8f72-f51ebcd036d9
Getting Started
SWE3030A - Software ...
Student Planner
Gmail
YouTube
Maps
Google Lens
News
Content
W3Schools Online We...
Other Bookmarks
Radar!
Home
Settings
About
Disclaimer
Build signature:
0000001a
Public Room
Connected Peers:
Peer 1
Peer 2
Your message...


Radar!
Home
Settings
About
Disclaimer
Build signature:
0000001a
Private Room
Connected Peers:
Peer 1
Peer 2
Room Password
Ensure all peers enter the same password.
CANCEL
SUBMIT
Your message...


Terminal
Local
Local (2)
Server running on port 5000
A user connected: _YX5N4lwG9cXcMAAAAAC
A user connected: 2cMV8B9Whv-DeCgEAAAD
undefined (2cMV8B9Whv-DeCgEAAAD) joined room: 5bd7ed44-43c3-4e27-8f72-f51ebcd036d9
undefined (2cMV8B9Whv-DeCgEAAAD) joined room: 5bd7ed44-43c3-4e27-8f72-f51ebcd036d9
undefined (_YX5N4lwG9cXcMAAAAAC) joined room: 2318bf2a-1464-4121-ad86-551fedfbb76c
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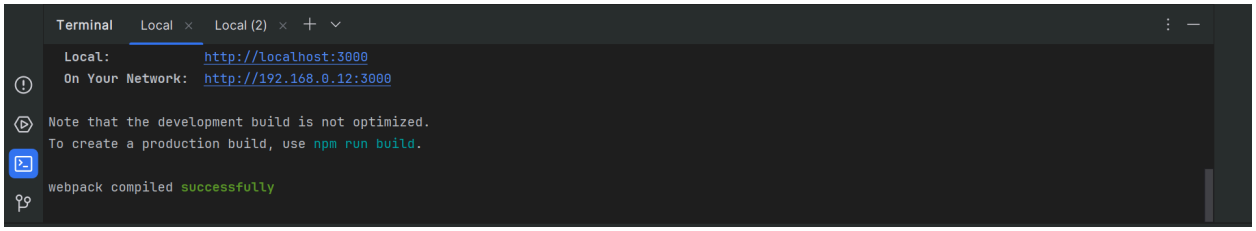
Terminal
Local
Local (2)
Local: http://localhost:3000
On Your Network: http://192.168.0.12:3000
Note that the development build is not optimized.
To create a production build, use npm run build.
webpack compiled successfully

4.2 Challenges Faced

- **Scalability:** Managing increased user load in a decentralized network.
- **Security Threats:** Addressing potential vulnerabilities in P2P communication.

5. Conclusion & Future Work

Radar! Underground Activism Hub provides a privacy-focused, decentralized communication platform for activists. The project's progress demonstrates the feasibility of alternative, censorship-resistant communication solutions.

Next Milestones Work

- **Enhancing Moderation Tools:** Developing advanced moderation features.
- **User Experience Improvements:** Refining UI for better accessibility.
- **Scalability Solutions:** Implementing better support for growing user numbers.
- **Educational Initiatives:** Raising awareness about the benefits of decentralized platforms.